

Force

In physics, a **force** is a push or pull acting upon an object as a result of the object's interaction with another object.

It is an external agent capable of altering an object's state of rest or motion, causing it to accelerate, decelerate, or change direction.

Key Characteristics

- **Vector Quantity**

Forces have both magnitude (strength) and direction.

- **SI Unit**

The standard metric unit for force is the **Newton (N)**.

One Newton is the force required to accelerate a 1 kg mass at a rate of **1 m/s²**.

- **Formula**

Force is calculated using Newton's Second Law of Motion:

$$[$$

$$F = m a$$

$$]$$

where

(F) = force,

(m) = mass,

(a) = acceleration.

Common Types of Forces

- **Contact Forces**

Applied through physical contact between objects.

Examples: friction, tension, air resistance, and a physical push/pull.

- **Non-Contact Forces**

Act over a distance without physical contact.

Examples: gravitational, electromagnetic, and nuclear forces.

These notes capture the essential concepts of force, its characteristics, and the primary categories of forces encountered in physics.